MySQL Stored Procedure – Assignment

Q1. Check Even or Odd

Create a stored procedure check_even_odd that accepts an integer and displays whether the number is Even or Odd.

Test cases:

```
CALL check_even_odd(25);
CALL check_even_odd(40);
```

Q2. Calculate Simple Interest

Create a stored procedure calculate_interest that accepts principal amount, rate of interest, and time in years and calculates Simple Interest.

Simple Interest = (Principal × Rate × Time) / 100

Test cases:

```
CALL calculate_interest(10000, 5, 2);
CALL calculate_interest(25000, 7, 3);
```

Q3. Calculate Grade

Create a stored procedure calculate_grade that accepts student ID, student name, and marks and displays the student's grade.

Marks	Grade
90–100	A+
80–89	A
70–79	B
60–69	C
50–59	D
Below 50	F

Test cases:

```
CALL calculate_grade(101, 'Amit', 85);
CALL calculate_grade(102, 'Priya', 72);
CALL calculate_grade(103, 'Rahul', 45);
```

Q4. Calculate Employee Bonus

Create a stored procedure calculate_bonus that accepts employee name and salary and calculates the bonus according to the following rules. Display employee name, salary, bonus percentage, bonus amount, and total salary.

Salary	Bonus
≥ 100000	20%
≥ 50000	15%
≥ 30000	10%
Below 30000	5%

Test cases:

```
CALL calculate_bonus('Amit', 120000);
CALL calculate_bonus('Priya', 65000);
CALL calculate_bonus('Rahul', 25000);
```

Q5. Find Maximum of Three Numbers —

Create a stored procedure find_maximum that accepts three integers and displays the largest number.

Test cases:

```
CALL find_maximum(25, 78, 45);
CALL find_maximum(100, 50, 75);
CALL find_maximum(30, 30, 20);
```

Q6. Employee Salary Analysis

Create a stored procedure salary_analysis that accepts employee name and salary. Determine the salary category and allowance, then calculate the final salary. Display employee name, salary, category, allowance percentage, allowance amount, and final salary.

Salary	Category	Allowance
≥ 100000	High	20%
≥ 50000	Medium	15%
Below 50000	Low	10%

Allowance = Salary × Allowance Percentage / 100

Final Salary = Salary + Allowance

Test cases:

```
CALL salary_analysis('Amit', 120000);
CALL salary_analysis('Priya', 60000);
CALL salary_analysis('Rahul', 35000);
```

Q7. Electricity Bill Calculation

Create a stored procedure calculate_electricity_bill that accepts consumer name and units consumed. Calculate the electricity bill using the slab-wise rates below. Display consumer name, units consumed, and total bill amount.

Units	Rate
First 100 units	₹2 per unit
Next 100 units	₹3 per unit
Next 200 units	₹5 per unit

Above 400 units	₹7 per unit

Test cases:

```
CALL calculate_electricity_bill('Amit', 80);
CALL calculate_electricity_bill('Priya', 150);
CALL calculate_electricity_bill('Rahul', 350);
CALL calculate_electricity_bill('Sneha', 500);
```